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Motivation

- **Objective:** Develop a high-performance racing bicycle through reverse engineering methodologies and computational analysis.
- **Utilization of computational software package:** Leveraging cutting-edge software tools for CAD (Computer-Aided Design), FEA (Finite Element Analysis), and CFD (Computational Fluid Dynamics) simulations.
- **Emphasis on performance optimization:** Focus on static loading analysis, material selection, cross sectional geometry design, and bicycle frame geometry to enhance speed, agility, and ergonomics.
- **Interdisciplinary approach:** Collaboration between mechanical engineers, material science experts, and cyclists to ensure a comprehensive design process.
- **Integration of real-world constraints:** Consideration of DFMA, material properties, cost-effectiveness, and complying regulatory standards to ensure an universal bicycle design.
- **Potential impact:** Enhancing competitiveness in professional racing circuits, improving recreational cycling experiences, and contributing to advancements in bicycle frame design.

Introduction

- **Research and Requirements Gathering:**
 - Define performance criteria (power transmission, material strength, weight).
 - Study existing racing bicycles and conduct research on current methods of evaluating performance.
- **Conceptual Design:**
 - Brainstorm design concepts considering aerodynamics, materials, and ergonomics.
 - Sketch and evaluate initial concepts based on performance criteria.
 - Measure and compare performance metrics between designs.
- **Computational Simulation:**
 - Select appropriate computational software packages (e.g., Ansys Workbench, Mechanical).
 - Create 3D models of the bicycle components and system.
 - Perform simulations to analyze aerodynamics, stress distribution, and performance characteristics.
- **Optimization:**
 - Identify areas for improvement based on simulation results.
 - Adjust design parameters such as frame geometry, material selection, and component placement.
 - Iterate simulations to refine the design and achieve optimal performance.
- **Prototyping and Testing:**
 - Construct prototypes based on optimized design.
 - Conduct real-world testing to validate performance metrics.
- **Final Design Documentation:**
 - Document final design specifications: materials, dimensions, and performance data.
 - SolidWork drawings & Frame design ANSYS simulations
- **Presentation and Evaluation:**
 - Present the final design to stakeholders, highlighting performance improvements and innovative features.

Materials

- **Material Selection:** Material choice selection dependent on performance, durability, and weight of the bicycle.
- **Prototype Validation:** Frame testing with prototype materials provides real-world data for refining design and identifying improvement areas.
- **Risk Mitigation:** Reduces the risk of product failure and ensures compliance with safety standards.

Table 1: Mechanical properties of the testing materials [1]

Materials	Modulus of Elasticity (GPa)	Poisson's Ratio	Density (Kg/M3)
Steel	205	0.29	7800
Aluminum 6061 T6	72	0.33	2700
Titanium Grade 9	95	0.30	4480
CFRP	415	0.10	1800

Methodology

Table 2: Loading Conditions performed on bike frame.

Loading Conditions	Description
Static Startup	Load of 700 N and 200 N applied to the top of the seat tube and on the bottom of the bracket
Steady State Pedaling	Load of 1000 N and 200 N applied to the bottom tube and head tube respectively
Vertical Impact	Load of 2250 N applied to the top of the bracket
Horizontal Impact	Load of 2250 N applied to the head tube horizontally
Rear Wheel Braking	Load of 750 N applied to the rear wheel bracket

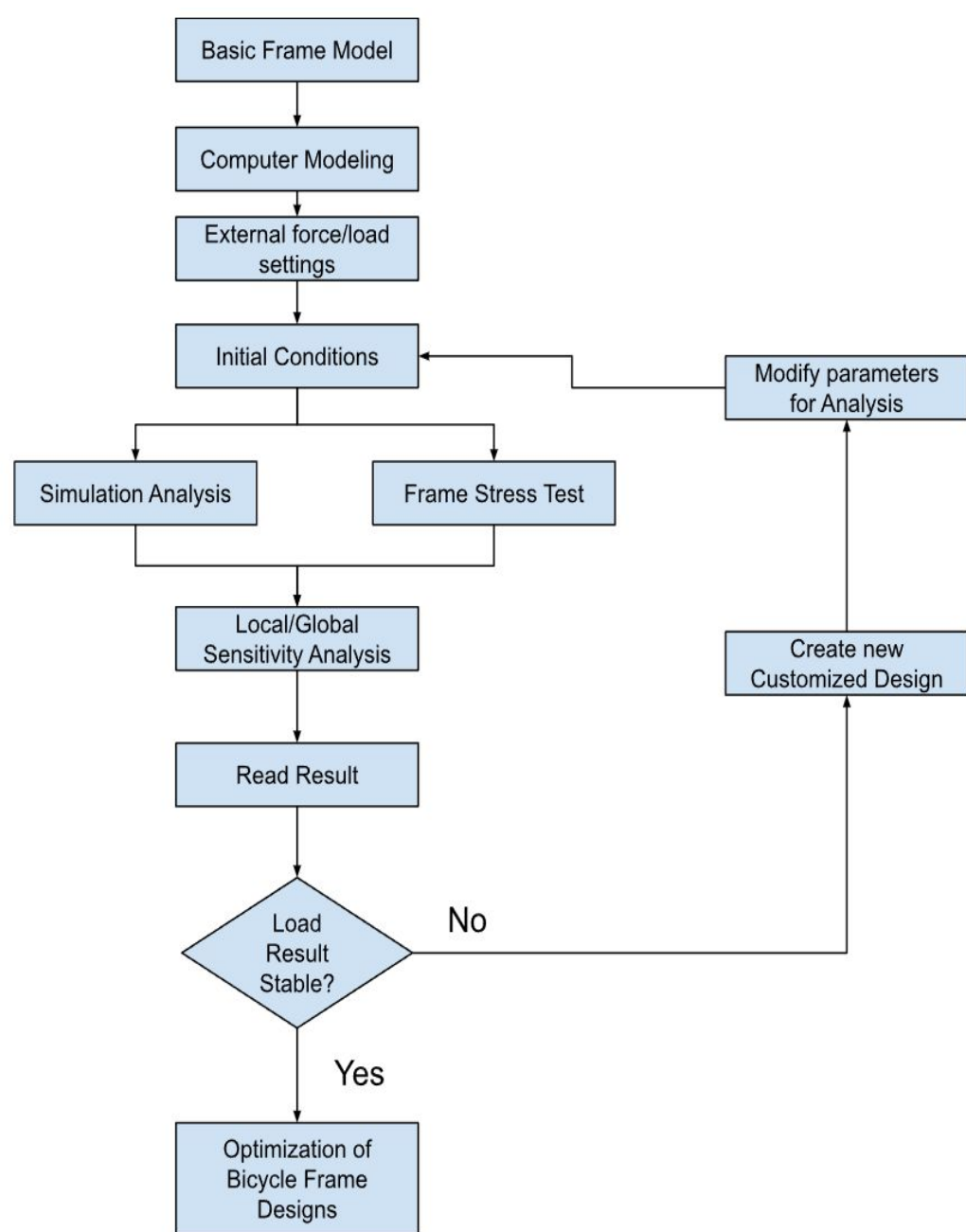


Figure 1: Flowchart of the Iterative Design Procedure for using Computer Aided Design into modeling the Final Design of the Prototype.

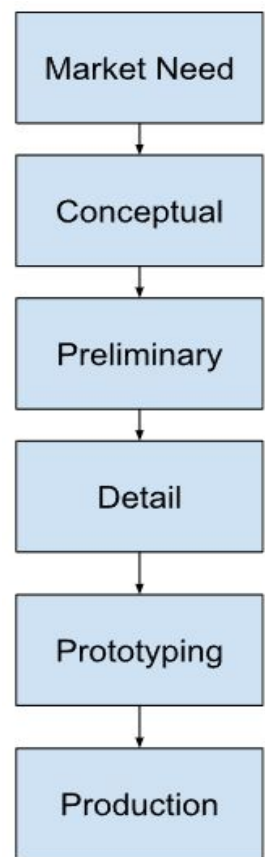


Figure 2: Design Requirements Flowchart

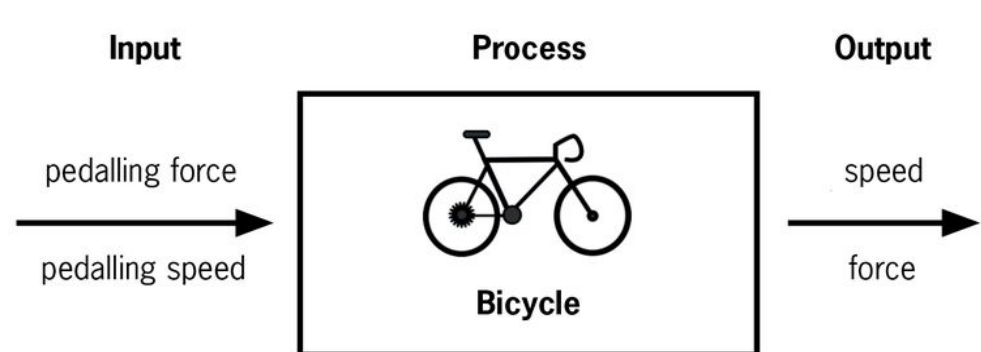


Figure 3: System Design Block Chart

Results

- **Steel**
 - Effective initial force conditions
 - Maintains structural integrity during normal cycling.
 - Potential for deformation; consider enhanced design or protection.
- **Aluminum 6061 T6:**
 - Similar stress values to steel across all conditions.
 - Comparable performance and maintenance needs.
- **Titanium Grade 9:**
 - Slightly varied stresses.
 - Better static startup and horizontal impact resistance.
 - Reduced maintenance requirements.
- **Carbon Fiber (CFRP):**
 - Generally lower stresses.
 - Less deformation and damage under similar loads.
 - Establish inspection protocols for vertical and horizontal impacts due to brittleness.

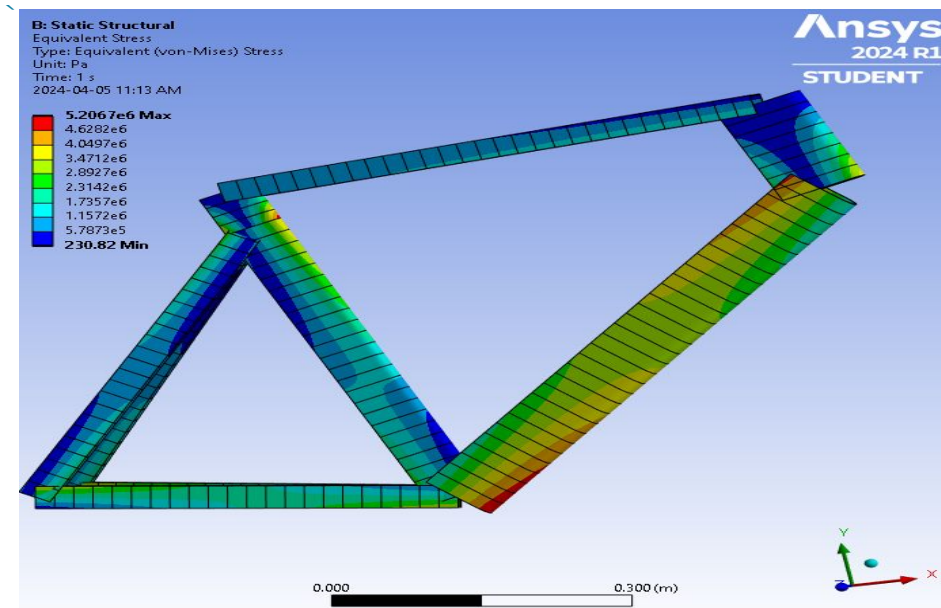


Figure 3: Frame design structural analysis on ANSYS.

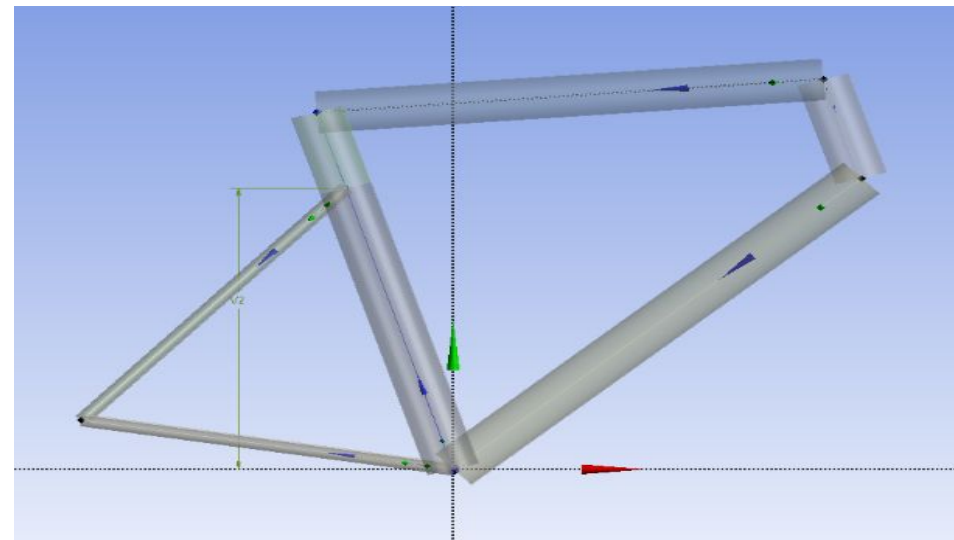


Figure 4: Final frame model constructed on ANSYS.



Figure 5: Final Bicycle Prototype Design.

Table 3: Simulation Results of Deformation (mm) for different materials

	Maximum Total Deformation (mm)			
Conditions	Steel	Aluminum 6061 T6	Titanium Grade 9	Carbon Fiber
Static Startup	0.0081404	0.023512	0.017838	0.0040985
Steady State Pedaling	0.0084963	0.024481	0.018578	0.0042668
Vertical Impact	0.0024997	0.0072276	0.0054793	0.012585
Horizontal Impact	0.030902	0.090021	0.068393	0.015749
Rear Wheel Braking	0.18428	0.52604	0.39793	0.089395

Conclusion

- **Project conclusion:**
 - Addresses challenges of racing bicycle design.
 - Addressed key performance metrics: Stress, Deflection, weight, geometry and strength
- **Improvements in Bike Components:**
 - Unique bicycle frame design with varying cross sections
 - Improvement in Stiffness-to-weight ratio,
 - Pedalling efficiency
- **Design Process**
 - Particular attention to the design of the bicycle frame cross sections
 - Iterative design iterations using ANSYS, fine-tune the bicycle's geometry and material choices to maximize performance while ensuring safety and comfort for the rider.

Recommendations

- **Computational Analysis:**
 - Dynamic analysis: Evaluate heat dissipation in critical components like the brakes and drivetrain through exposure to velocities.
 - Vibration analysis: Assess the bicycle's ride comfort, durability, and stress by simulating vibrations caused by road irregularities.
- **Validation and Testing:**
 - Fabricate prototypes: Build physical prototypes based on the optimized design to validate simulation results and verify performance.
 - Conduct wind tunnel tests: Validate aerodynamic predictions by testing scaled models in a wind tunnel and comparing results with computational simulations.
 - Perform mechanical testing: Subject prototypes to standardized tests, such as ASTM standards for bicycles, to assess their structural integrity and compliance with safety regulations.
- **Iterative Improvement:**
 - Collect feedback from test riders and professional cyclists to identify areas for improvement and refinement.
 - Additional iterations with ANSYS and implement additional DFMA methodologies.

References

- [1] S. P. A. Deepak, N. S. Daniel, H. H. and Jinuchandran, "Stress Analysis of Bicycle Frame using Different Materials by FEA," GRD Journal for Engineering, vol. 6, no. 7, Jun. 2021.
- [2] A. G. Kitselis, C. S. Nikolakea, and D. E. Manolakos, "The design process of an optimized road racing bicycle frame," Machines, vol. 10, no. 2, p. 149, Feb. 2022. doi:10.3390/machines10020149
- [3] Current Perspectives of Cross-Country Mountain Biking: Physiological and Mechanical Aspects, Evolution of Bikes, Accidents and Injuries. Available: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9565958/>